1 Week 9

Let m be a set function defined for all sets in a σ-algebra $\mathcal{A}$ with values in $[0,\infty]$. Assume m is countably additive over countable disjoint collections of sets in $\mathcal{A}$.

Question 1.1 (Royden 2.1)**.** Prove that if A and B are two sets in $\mathcal{A}$ with $A \subseteq B$, then $m(A) \leq m(B)$. This property is called *monotonicity*.

Proof. $\square$

Question 1.2 (Royden 2.2)**.** Prove that if there is a set A in the collection $\mathcal{A}$ for which $m(A) < \infty$ then $m(\varnothing) = 0$.

Proof. $\square$

Question 1.3 (Royden 2.3)**.** Let $\{E_k\}_{k=1}^{\infty}$ be a countable collection of sets in $\mathcal{A}$. Prove that $m(\cup_{k=1}^{\infty} E_k) \leq \sum_{k=1}^{\infty} m(E_k)$.

Proof. $\square$

Question 1.4 (Royden 2.4)**.** A set function c, defined on all subsets of $\mathbb{R}$ is defined as follows. Define $c(E)$ to be ∞ if E has infinitely many members and $c(E)$ to be the number of elements in E if E is finite; define $c(\varnothing) = 0$. Show that c is a countably additive and translation invariant set function. This set function is called the **counting measure**.

Proof. $\square$

Question 1.5 (Royden 2.8)**.** Let B be the set of rational numbers in the interval $[0,1]$, and let $\{I\}_{k=1}^{\infty}$ be a finite collection of open intervals that covers B. Prove that $\sum_{k=1}^{n} m^*(I_k) > 1$.

Proof. $\square$

Question 1.6 (Royden 2.10)**.** Let A and B be bounded sets for which there is an $\alpha > 0$ such that $|a-b| > \alpha$ for all $a \in A, b \in B$. Prove that $m^*(A \cup B) = m^*(A) + m^*(B)$.

Proof. $\square$

Question 1.7 (Royden 2.11)**.** Prove that if a σ-algebra of subsets of $\mathbb{R}$ contains intervals of the form (a,∞), then it contains all intervals.

Proof. $\square$

Question 1.8 (Royden 2.19)**.** Let E have finite outer measure. Show that if E is not measurable, then there is an open set $\mathcal{O}$ containing E that has finite outer measure and for which

$$m^*(\mathcal{O} \setminus E) > m^*(\mathcal{O}) - m^*(E).$$

Proof. $\square$

Question 1.9 (Royden 2.20: Lebesgue)**.** Let E have finite outer measure. Show that E is measurable if and only if for each open, bounded interval (a,b),

$$b - a = m^*((a,b) \cap E) + m^*((a,b) \setminus E).$$

Proof. $\square$

Question 1.10 (Royden 2.24)**.** Show that if E_1 and E_2 are measurable, then

$$m(E_1 \cup E_2) + m(E_1 \cap E_2) = m(E_1) + m(E_2).$$

Proof. □

Question 1.11 (Royden 2.25)**.** Show that the assumption that $m(B_1) < \infty$ is necessary in part (ii) of the theorem regarding continuity of measure.

Proof. □

Question 1.12 (Royden 2.27)**.** Let $\mathcal{M}'$ be any σ-algebra of subsets of $\mathbb{R}$ and m' a set function on $\mathcal{M}'$ which takes values in $[0, \infty]$, is countably additive, and such that $m'(\varnothing) = 0$.

(i) Show that m' is finitely additive, monotone, countably monotone, and possesses the excision property.

(ii) Show that m' possesses the same continuity properties as Lebesgue measure.

Proof. □

Question 1.13 (Royden 2.28)**.** Show that continuity of measure together with finite additivity of measure implies countable additivity of measure.

Proof. □